

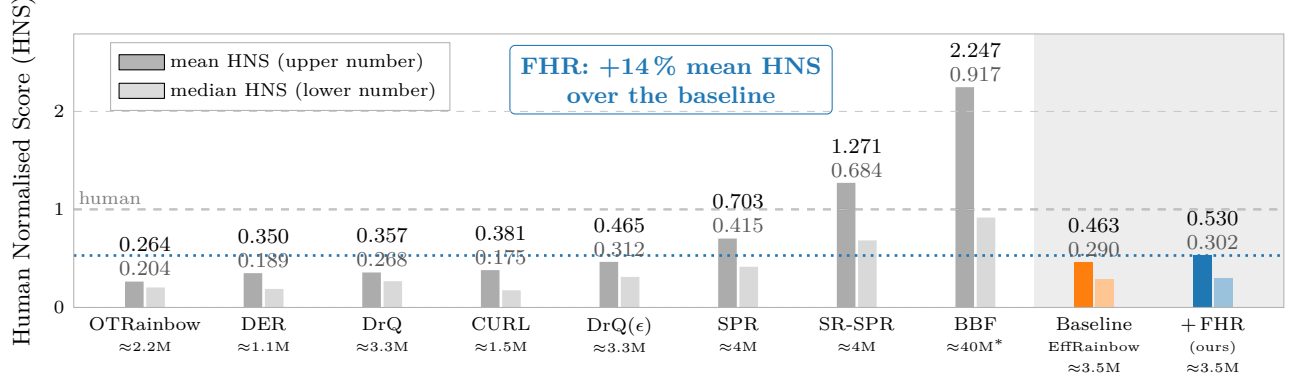


Figure 1: Atari-100k, 26 games, 100k environment steps: mean and median human-normalised score of published model-free agents (grey; the counts under the names are the trainable parameters of each paper’s network) and of our EfficientRainbow baseline with and without the FHR loss (270 runs pooled, seeds 0–4). Published rows are quoted from one source each (EfficientZero Table 1; BBF Table A.1).

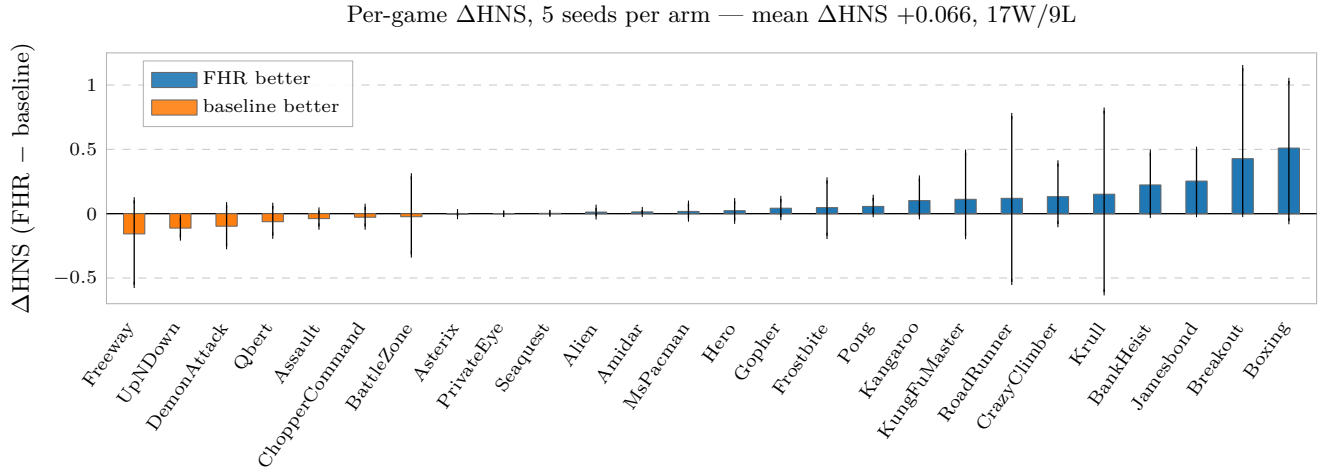


Figure 2: Per-game difference in human-normalised score between the FHR arm and the baseline (bar: difference of the per-game mean scores over the seeds; whisker: 95 % percentile interval of an unpaired seed bootstrap within the game, 2000 replicates). Positive bars favour FHR.

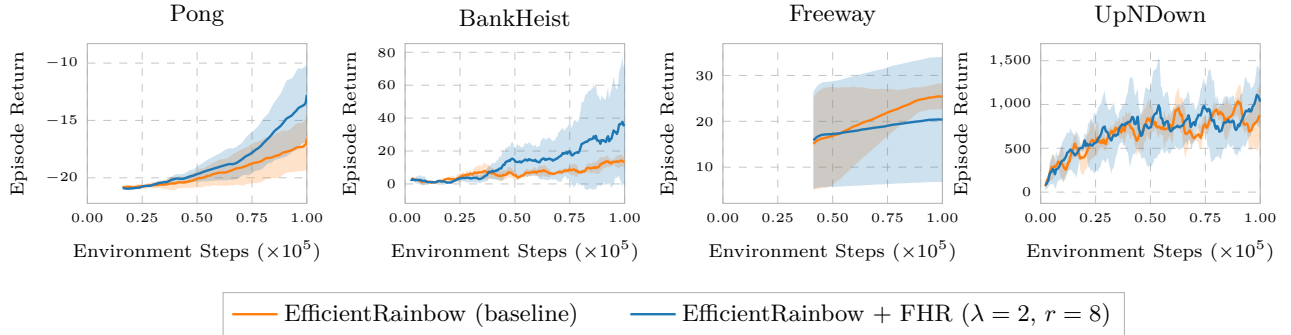


Figure 3: Training curves on Pong, BankHeist, Freeway and UpNDown: EfficientRainbow baseline and the same agent with the FHR loss. Lines: mean over 4 seeds of the raw episode return smoothed with a moving average of 20 episodes, on a common grid of 200 points over the 100k environment steps; shaded: $\pm$ one standard deviation across seeds. All 26 games are in Figures 4–5.

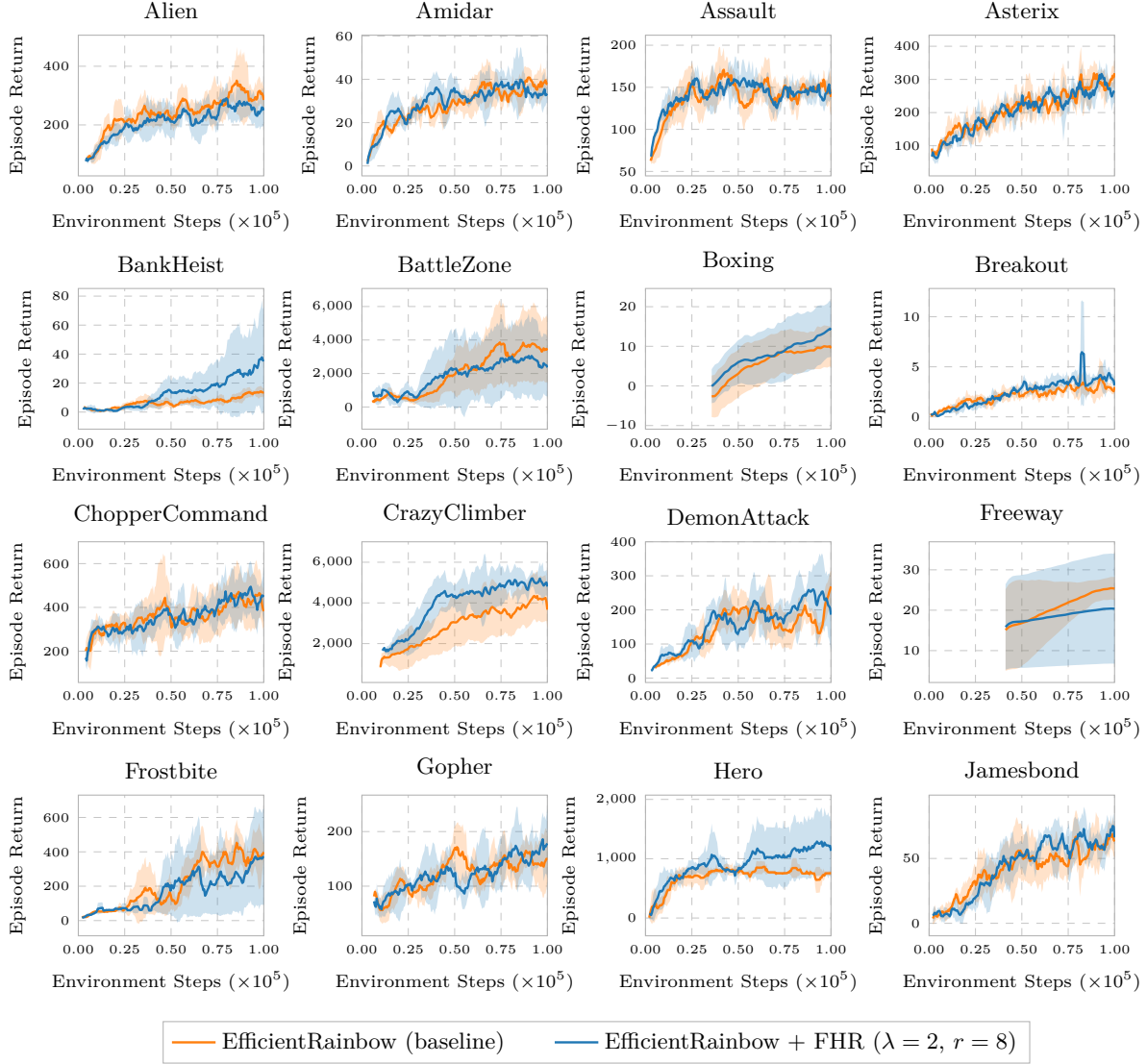


Figure 4: Training curves on the 26 Atari-100k games: EfficientRainbow baseline and the same agent with the FHR loss. Lines: mean over 4 seeds of the raw episode return smoothed with a moving average of 20 episodes, on a common grid of 200 points over the 100k environment steps; shaded: $\pm$ one standard deviation across seeds. Panels whose FHR arm has fewer seeds say so in the title. Games Alien–Jamesbond.

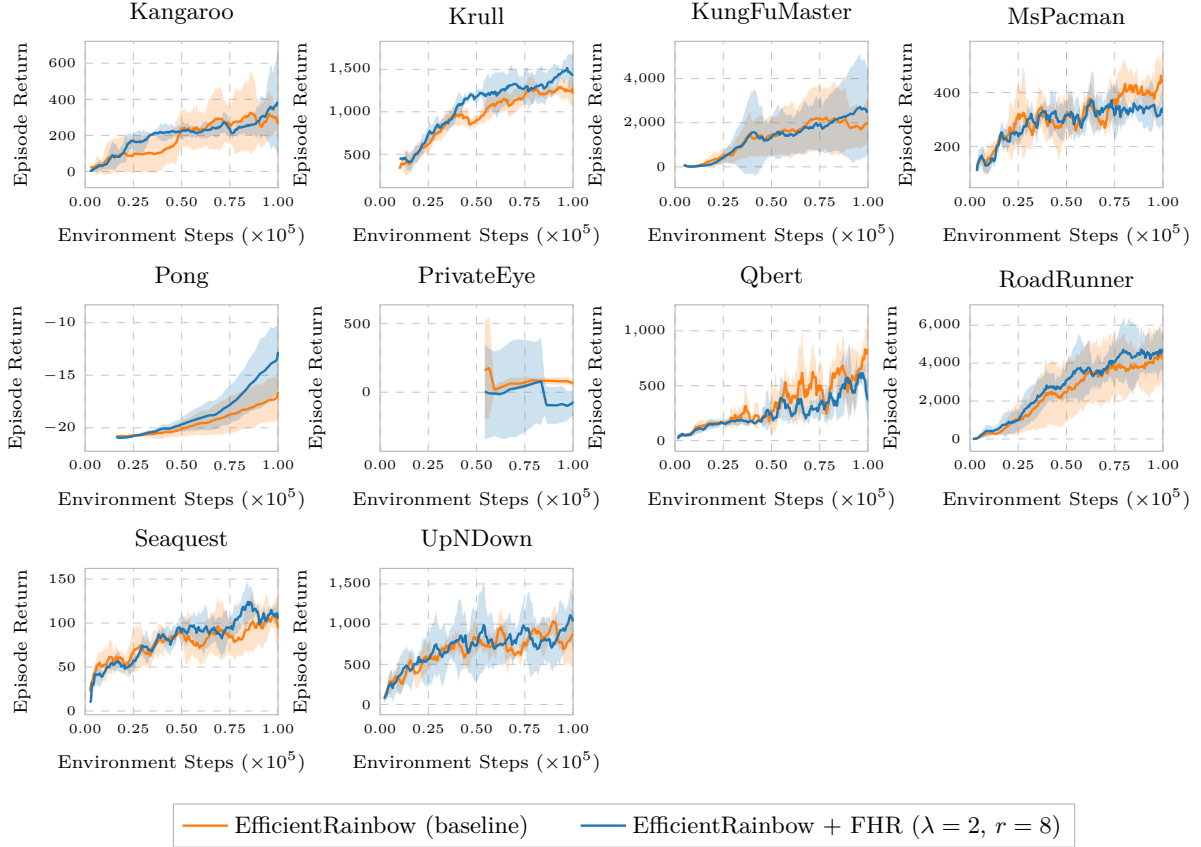


Figure 5: Training curves on the 26 Atari-100k games (continued): EfficientRainbow baseline and the same agent with the FHR loss. Lines: mean over 4 seeds of the raw episode return smoothed with a moving average of 20 episodes, on a common grid of 200 points over the 100k environment steps; shaded: $\pm$ one standard deviation across seeds. Panels whose FHR arm has fewer seeds say so in the title. Games Kangaroo–UpNDown.